



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 15.12.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 14.12.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 14-दिसंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 14.12.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 15-Dec-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs)	56805	61762	39343	21377	2666	181953
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1222	1463	1021	418	48	4172
Hydro Gen (MU)	115	40	67	16	14	252
Wind Gen (MU)	13	153	73	-	-	239
Solar Gen (MU)*	165.90	104.34	90.32	3.49	1.11	365
Energy Shortage (MU)	1.51	0.00	0.00	0.00	0.00	1.51
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	66783	73474	50072	22003	2819	210497
Time Of Maximum Demand Met	10:44	09:56	09:28	17:49	17:32	09:56

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.046	0.00	0.42	7.24	7.66	77.27	15.07

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	9511	0	165.7	74.1	-1.0	667	0.00
	Haryana	8830	0	162.3	106.8	-0.4	413	0.00
	Rajasthan	17890	0	330.4	114.5	-0.1	494	0.00
	Delhi	4306	0	73.6	73.3	-1.2	321	0.00
	UP	19562	0	333.7	113.6	-0.3	494	0.00
	Uttarakhand	2419	0	44.8	33.7	0.1	81	0.08
	HP	2146	0	39.4	32.1	-3.0	81	0.00
	J&K(UT) & Ladakh(UT)	3076	0	63.9	58.7	0.2	624	1.43
	Chandigarh	229	0	4.0	3.9	0.1	37	0.00
WR	Railways_NR ISTS	206	0	3.9	3.8	0.1	38	0.00
	Chhattisgarh	4660	0	96.0	34.8	0.2	339	0.00
	Gujarat	21325	0	403.8	167.9	-0.8	557	0.00
	MP	18221	0	335.0	199.9	-2.7	909	0.00
	Maharashtra	26376	0	551.9	192.0	-6.0	473	0.00
	Goa	719	0	14.8	14.2	0.2	48	0.00
	DNHDDPDCL	1301	0	29.8	30.0	-0.2	50	0.00
	AMNSIL	801	0	17.6	8.3	0.4	283	0.00
	BALCO	528	0	12.6	12.7	-0.1	8	0.00
SR	RIL JAMNAGAR	76	0	1.6	1.6	0.0	0	0.00
	Andhra Pradesh	9345	0	190.7	52.3	-1.0	517	0.00
	Telangana	12653	0	237.9	142.6	-0.4	1053	0.00
	Karnataka	12488	0	223.6	50.8	-3.7	491	0.00
	Kerala	4015	0	80.5	65.6	0.6	198	0.00
	Tamil Nadu	14384	0	280.1	161.1	-2.2	746	0.00
	Puducherry	399	0	8.5	8.4	0.0	71	0.00
ER	Bihar	4970	0	88.6	77.7	-1.0	266	0.00
	DVC	3182	0	62.2	-44.2	-6.3	331	0.00
	Jharkhand	2054	0	36.7	26.1	-0.4	304	0.00
	Odisha	4939	0	91.4	16.1	-1.9	466	0.00
	West Bengal	6962	0	137.4	11.9	-2.0	241	0.00
	Sikkim	119	0	1.8	1.7	0.0	50	0.00
	Railways_ER ISTS	21	0	0.2	0.1	0.0	0	0.00
NER	Arunachal Pradesh	159	0	2.8	2.8	-0.2	14	0.00
	Assam	1553	0	26.8	19.4	0.3	131	0.00
	Manipur	246	0	3.9	3.9	0.0	50	0.00
	Meghalaya	384	0	6.6	5.9	-0.3	126	0.00
	Mizoram	152	0	2.4	1.8	-0.1	10	0.00
	Nagaland	173	0	2.5	2.4	0.0	27	0.00
	Tripura	225	0	2.6	3.7	-0.7	44	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-5.6	-2.7	-12.8	-13.5
Day Peak (MW)	-522.0	-153.0	-547.0	-771.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	232.6	-200.7	164.3	-202.7	6.5	0.0
Actual(MU)	231.3	-203.9	157.7	-197.7	4.9	-7.6
O/D/U/D(MU)	-1.2	-3.2	-6.7	5.0	-1.6	-7.6

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	4379	10909	8358	4338	518	28502	48
State Sector	7733	12849	7549	1909	248	30287	52
Total	12112	23758	15907	6247	765	58789	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	683	1398	570	674	9	3335	74
Lignite	23	7	27	0	0	56	1
Hydro	115	40	67	16	14	252	6
Nuclear	32	68	70	0	0	170	4
Gas, Naptha & Diesel	3	11	4	0	24	42	1
RES (Wind, Solar, Biomass & Others)	205	262	191	7	1	665	15
Total	1061	1785	929	697	48	4519	100

Share of RES in total generation (%)	19.30	14.65	20.57	0.94	2.34	14.71
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	33.16	20.67	35.36	3.20	30.90	24.04

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.022
Based on State Max Demands	1.048

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	210497	9:56	545
Non-Solar hr	195448	18:19	0

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 15-Dec-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	249	0.0	6.2	-6.2
3	765 kV	GAYA-VARANASI	2	0	1220	0.0	18.9	-18.9
4	765 kV	SASARAM-FATEHPUR	1	0	508	0.0	7.4	-7.4
5	765 kV	GAYA-BALIA	1	0	776	0.0	13.4	-13.4
6	400 kV	PUSAULI-VARANASI	1	0	214	0.0	4.0	-4.0
7	400 kV	PUSAULI -ALLAHABAD	1	0	117	0.0	2.0	-2.0
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	643	0.0	9.5	-9.5
9	400 kV	PATNA-BALIA	2	0	1265	0.0	21.4	-21.4
10	400 kV	NAUBATPUR-BALIA	2	0	360	0.0	3.6	-3.6
11	400 kV	BIHARSHARIFF-BALIA	2	0	360	0.0	6.3	-6.3
12	400 kV	MOTIHARI-GORAKHPUR	2	0	591	0.0	10.0	-10.0
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	61	0.0	0.7	-0.7
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.6	103.3	-102.7
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	0	1062	0.0	16.0	-16.0
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	302	1154	0.0	4.7	-4.7
3	765 kV	JHARSUGUDA-DURG	2	0	770	0.0	13.6	-13.6
4	400 kV	JHARSUGUDA-RAIGARH	4	0	844	0.0	10.5	-10.5
5	400 kV	RANCHI-SIPAT	2	1	0	0.0	0.0	0.0
6	220 kV	BUDHIPADAR-RAIGARH	1	0	101	0.0	1.4	-1.4
7	220 kV	BUDHIPADAR-KORBA	2	103	130	0.1	0.0	0.1
ER-WR						0.1	46.1	-46.1
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	260	0.0	6.2	-6.2
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1645	0.0	39.7	-39.7
3	765 kV	ANGUL-SRIKAKULAM	2	0	2663	0.0	52.7	-52.7
4	400 kV	TALCHER-I/C	2	167	614	0.0	0.4	-0.4
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	98.6	-98.6
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	301	0	2.9	0.0	2.9
2	400 kV	ALIPURDUAR-BONGAIGAON	2	492	0	6.1	0.0	6.1
3	220 kV	ALIPURDUAR-SALAKATI	2	91	0	1.3	0.0	1.3
ER-NER						10.3	0.0	10.3
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	667	0	15.8	0.0	15.8
NER-NR						15.8	0.0	15.8
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	2023	0.0	36.2	-36.2
2	HVDC	VINDHYACHAL B/B	-	0	55	0.0	1.2	-1.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1457	0.0	30.8	-30.8
4	765 kV	GWALIOR-AGRA	2	0	1864	0.0	27.7	-27.7
5	765 kV	GWALIOR-PHAGI	2	596	1563	1.4	19.0	-17.6
6	765 kV	JABALPUR-ORAI	2	0	904	0.0	28.4	-28.4
7	765 kV	GWALIOR-ORAI	1	879	0	12.9	0.0	12.9
8	765 kV	SATNA-ORAI	1	0	905	0.0	17.9	-17.9
9	765 kV	BANASKANTHA-CHITORGARH	2	1239	852	6.2	7.4	-1.2
10	765 kV	VINDHYACHAL-VARANASI	2	89	1615	0.0	25.1	-25.1
11	400 kV	ZERDA-KANKROLI	1	320	128	1.7	0.8	0.9
12	400 kV	ZERDA -BHINMAL	1	221	89	1.2	0.6	0.7
13	400 kV	VINDHYACHAL -RIHAND	1	967	0	0.0	22.3	-22.3
14	400 kV	RAPP-SHUJALPUR	2	598	249	3.3	1.1	2.2
15	400 kV	NEEMUCH-Chittorgarh	2	604	346	3.5	1.4	2.1
16	220 kV	BHANPURA-RANPUR	1	0	192	0.0	3.2	-3.2
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	1.9	-1.9
18	220 kV	MEHGAON-AURAIYA	1	106	0	1.1	0.0	1.1
19	220 kV	MALANPUR-AURAIYA	1	70	6	0.6	0.0	0.5
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						31.9	225.0	-193.1
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1007	0.0	24.0	-24.0
2	HVDC	RAIGARH-PUGALUR	2	0	2505	0.0	46.8	-46.8
3	765 kV	SOLAPUR-RAICHUR	2	1006	622	6.2	3.0	3.1
4	765 kV	WARDHA-NIZAMABAD	2	0	1810	0.0	28.8	-28.8
5	765 kV	WARORA-WARANGAL(NEW)	2	0	1939	0.0	32.7	-32.7
6	400 kV	KOLHAPUR-KUDGI	2	1456	0	25.5	0.0	25.5
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	135	2.5	0.0	2.5
WR-SR						34.1	135.3	-101.2

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	-130	112	-2	-0.04	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	324	-185	-15	-0.36	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-248	-129	-195	-4.69	
	NER	132kV GELEPHU-SALAKATI	-26	-14	-20	-0.48	
	NER	132kV MOTANGA-RANGIA	-7	0	-1	-0.03	
NEPAL	NR	NEPAL IMPORT (FROM UP)	-65	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-65	0	-39	-0.94	
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-153	41	-74	-1.78	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-488	0	-487	-11.70	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-771	-405	-563	-13.50	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-59	0	-47	-1.12	

CROSS BORDER EXCHANGE SCHEDULE

Date of Reporting: 15-Dec-2024

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	0.00	0.00	6.92	0.00	0.00	0.00	0.00	0.00	6.92	
Nepal	0.13	0.00	0.68	0.00	0.00	0.00	1.34	0.00	0.00	2.15
Bangladesh	12.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.15
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	12.28	0.00	7.60	0.00	0.00	0.00	1.34	0.00	0.00	21.22

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.66
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.66

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.66	0.00	-6.92	0.00	0.00	0.00	0.00	0.00	-6.26
Nepal	-0.13	0.00	-0.68	0.00	0.00	-1.34	0.00	0.00	-2.15
Bangladesh	-12.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-12.15
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-11.62	0.00	-7.60	0.00	0.00	-1.34	0.00	0.00	-20.56

Date of Reporting: 15-Dec-2024

15 Min (INSTANTANEOUS) ALL INDIA GRID FREQUENCY, GENERATION & DEMAND MET (SCADA DATA)												
TIME	FREQUENCY (Hz)	DEMAND MET (MW)	NUCLEAR (MW)	WIND (MW)	SOLAR (MW)	HYDRO** (MW)	GAS (MW)	THERMAL (MW)	OTHERS* (MW)	NET DEMAND MET (MW)	TOTAL GENERATION (MW)	NET TRANSNATIONAL EXCHANGE (MW)
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I=A-(C+D))	(J=B+C+D+E+H+G+I)	(K)
0:00	50.00	144122	6559	14498	4675	6746	1738	115058	2982	124949	152256	-876
0:15	49.95	142893	6560	14535	4674	6519	1741	114155	2999	123684	151183	-966
0:30	50.00	142468	6556	14326	4649	6890	1726	113145	2980	123493	150272	-989
0:45	49.95	142109	6570	14190	4649	6568	1727	113243	2983	123270	149930	-943
1:00	50.00	140747	6562	14090	4648	6340	1731	113231	2985	122009	148677	-1037
1:15	50.00	139364	6568	14023	4645	6247	1718	111528	2989	120696	147718	-1130
1:30	50.00	138753	6565	13954	4645	6131	1717	111000	2985	120154	146997	-1098
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